FIT5147 Data Exploration and Visualisation
Programming Exercise 2: R Shiny

Reflection and Justification of your PE2 Visualisation Design

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Section: VIS 1

A grouped bar chart was selected to display the proportion of observations per Australian season, broken down across four species. This idiom was preferred over a stacked bar chart because it supports direct, fair comparison — each bar originates from a shared zero baseline, eliminating the cognitive burden of mentally subtracting lower layers to compare upper ones. This aligns with Bertin's principle that position along a common scale is the most accurately decoded visual variable, making grouped bars the strongest choice when cross-category comparison is the primary analytical task (Bertin, 1983).

Colour hue is mapped to species using the Dark2 ColorBrewer palette, a qualitative scheme specifically designed for categorical data with no implied ordering. Dark2 is also relatively colourblind-friendly, important for an academic audience where accessibility cannot be assumed (Brewer, 2013). The y-axis encodes each species-season combination as a proportion of all observations in the dataset, preserving the true scale differences between species. This means the Orange-bellied Parrot's near-invisible bars honestly reflect its critically low representation in the dataset, rather than inflating it through within-species normalisation. The grouped structure still enables fair seasonal comparison within each species, as readers can track how a given species' proportional share shifts across seasons. The x-axis arranges seasons in Australian calendar order (Summer through Spring), providing temporal coherence. A minimal theme (theme_minimal) was applied to reduce chartjunk — vertical gridlines on the x-axis are suppressed (panel.grid.major.x) since no x-position reading is required, as seasons are labelled directly on the axis. Horizontal gridlines are retained to support reading proportions against the y-axis. The legend is positioned below the chart to avoid occluding the data area.

Section: Interactive MAP

A proportional symbol map was chosen over a choropleth because the data consists of point-based observations at specific coordinates rather than aggregated regional statistics. Mapping to administrative boundaries would misrepresent spatial precision and imply false uniformity within regions. Circle markers are positioned at exact observation coordinates, preserving geographic fidelity.

Colour hue encodes species identity using the same Dark2 palette as VIS 1, maintaining visual consistency across both components and reducing the cognitive load of learning a new encoding. Radius encodes observation count at each coordinate using a square-root scaling function (sqrt(obs_count) * 2.5), which compresses the large range of counts into a perceptually proportional range — a recognised technique to prevent dominant locations from visually overwhelming sparser ones (Dent et al., 2009). Semi-transparent fill (65% opacity) manages overlap without completely obscuring underlying markers. The CartoDB Positron basemap was deliberately chosen for its neutral grey palette, which ensures the species-coloured markers remain the visual figure against an unobtrusive ground, consistent with figure-ground theory in cartographic design (MacEachren, 1995).

Interactivity includes species checkboxes (default all selected) and season radio buttons (default Full Year), using leafletProxy() to update markers reactively without re-rendering the base map — minimizing server-side processing. Hover tooltips surface scientific name, common name, observation count, and state or territory, giving non-specialist users contextual information without cluttering the map surface.

Additional Reflections

One data anomaly warrants acknowledgement: a single record has a missing stateProvince value, recoded as "Not recorded" and retained per the specification's instruction not to clean the data. Records where observationDate could not be parsed carry an NA season value and are excluded from VIS 1 and seasonal map filtering, though they remain present in the Full Year view. The Orange-bellied Parrot's near-invisible markers reflect a data limitation — its critically endangered status — not a design failure. Ecological claims regarding migration and endangered status are drawn from BirdLife Australia (2024a, 2024b). Typography uses Georgia (serif) for headings and 13px sans-serif for body text.

AI Declaration

Prompts and purposes:

1. "What are the step-by-step requirements for this assignment and how should I approach building it?" — Used to plan the project structure and understand specification requirements.
2. "Which visual idiom best supports fair seasonal comparison for this dataset, and how should I justify it?" — Used to inform the choice of grouped bar chart over stacked bar for VIS 1.
3. "What is an appropriate colour palette for four categorical species in ggplot2 and Leaflet?" — Used to select and apply the Dark2 ColorBrewer palette consistently across both visualizations.
4. "Generate R Shiny code implementing the fixedPage layout, ggplot2 chart, and Leaflet proportional symbol map per the spec requirements." — Used to produce the initial application code, which was then tested and debugged iteratively.

5. "Draft the written reflection for VIS 1, MAP, and Additional Reflections sections using visualization theory." — Used to structure and generate content for the reflection document.

References

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- Brewer, C. A. (2013). *ColorBrewer 2.0: Color advice for cartography*. Retrieved from http://colorbrewer2.org
- Dent, B. D., Torguson, J., & Hodler, T. (2009). *Cartography: Thematic map design* (6th ed.). McGraw-Hill.
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